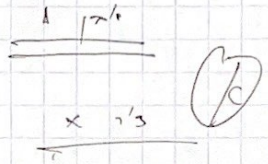
1.9



$$x = V_{0x} \cdot t$$

$$x = V_0 \cdot \cos\theta \cdot t$$

$$100 = 50 \cdot \cos 60^\circ \cdot t$$

$$100 = 25\,t$$

$$t = 4\,(s)$$

$$y = y_0 + V_{0y} \cdot t + \frac{1}{2} a t^2$$

$$y = y_0 + V_0 \sin\theta \cdot t + \frac{1}{2} a t^2$$

$$y = 0 + 50 \cdot \sin 60 \cdot 4 + \frac{1}{2}(-10) \cdot 4^2$$

$$y = 93.205\,m$$

$$y_1 = y_2$$

$$y_2 = y_0 + V_{0y} \cdot t + \frac{1}{2} a t^2$$

$$y_2 = 0 + V_{0_2} \cdot 4 - 5 \cdot 4^2$$

$$93.205 = 4V_0^2 - 80$$

$$V_{0x} = V_0 \cdot \cos\theta = 25\ \frac{m}{s}$$

$$V_{0y} = V_0 \cdot \sin\theta + at$$

$$V_{0y} = 50 \sin 60 - 10 \cdot 4 = 3.3\ \frac{m}{s}$$

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{25^2 + 3.3^2}$$

$$V = 25.217\ \frac{m}{s}$$

$$\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{25}{3.3}\right)$$

$$\theta = 82.43^\circ$$

$$\theta = 7.57^\circ$$

$$V_{02} = 43.3\ \frac{m}{s}$$

$$V_y = V_{0y} + at$$

$$V_y = 43.3 - 10 \cdot 4$$

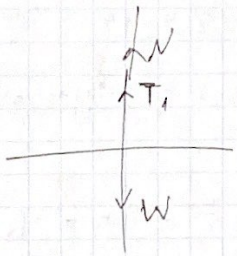
$$V_y = 3.3\ \frac{m}{s}$$

$V_0 = 8$

$a = 1$

$V = V_0 + at$

$V = 8 + 5 = 13$



$T_1 + N = 700$

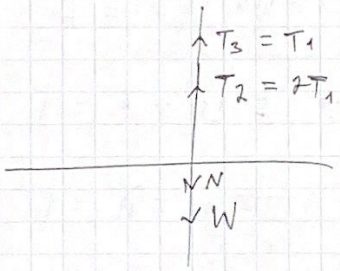
$T_2 + T_3 = 900$

$3T_1 = 3000$

$4T_1 = 900$

$T_1 = 225$

$T_1 + N = 700$



$3T_1 = 200 + N$

$T_1 = 700 - N$

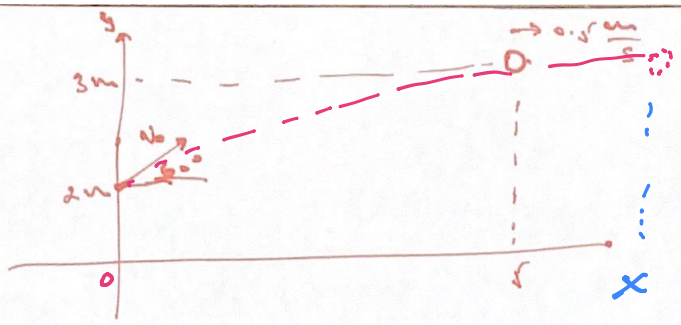
$3(700 - N) = 200 + N$

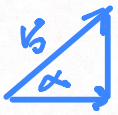
$2100 - 3N = 200 + N$

$4N = 2300$

$N = 575$

1.15





$x = x_0 + vt$

ב) נבחר בציר x כציר הנע במהירות קבועה ובציר y כציר הנע בתאוצה קבועה.

ציר x

$x = v_0 \cos 30° \cdot t$

$x = 0.5\sqrt{3}\, v_0 t$

$x_0 = 2\sqrt{3} + 0.5t$

$x = x_0$

$0.5\sqrt{3}\, v_0 t = 2\sqrt{3} + 0.5t$

$v_0 t = 10 + t$

ציר y

$y = y_0 + v_{0y} t - \frac{1}{2} g t^2$

$y_0 = 2$

$v_{0y} = v_0 \sin 30°$

$y = 2 + 0.5 v_0 t - 5t^2$

$3 = 2 + 0.5(10 + t) - 5t^2$

$3 = 2 + 5 + 0.5t - 5t^2$

$5t^2 - 0.5t - 5 = 0$

$t_1 = -1.154$

$t_2 = 1.327$

$v_0 t = 10 + t$

$v_0 \cdot 1.327 = 10 + 1.327$

$v_0 = \frac{11.327}{1.327}$

$v_0 = 8.535 \frac{m}{s}$

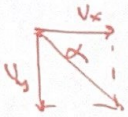
③

$v_x = v_0 \cos 60 = 4.268 \frac{m}{s}$

$v_y = v_{0y} - gt = 7.391 - 10 \cdot 1.327 = -5.878 \frac{m}{s}$

$v = \sqrt{v_x^2 + v_y^2}$

$v = 7.264 \frac{m}{s}$



$\alpha = \tan^{-1}\left(\frac{v_y}{v_x}\right)$

$\alpha = 54^\circ$

④

$d = v_0 \cos 60^\circ \cdot t$

$d = 4.268 \cdot 1.712$

$d = 7.31\,m$

$y = y_0 + v_{0y} t - \frac{1}{2} g t^2$

$0 = 2 + v_0 \cdot \sin 60^\circ \cdot t - 5 t^2$

$0 = 2 + 7.391 t - 5 t^2$

$t_1 = -0.366$

$t_2 = 1.712$

$v_x = v_0 \cos 60^\circ = 4.268 \frac{m}{s}$

$v_y = v_{0y} - 10t$

$v_y = v_0 \sin 60^\circ - 10 \cdot 1.712$

$v_y = 7.391 - 17.12$

$v_y = -9.729 \frac{m}{s}$

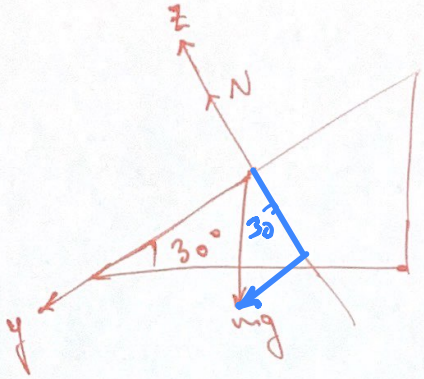
$v = \sqrt{v_x^2 + v_y^2}$

$v = 10.62 \frac{m}{s}$

$\alpha = \tan^{-1}\left(\frac{v_y}{v_x}\right)$

$\alpha = 66.31^\circ$

1.18



א. נבחר ציר y בכיוון התנועה

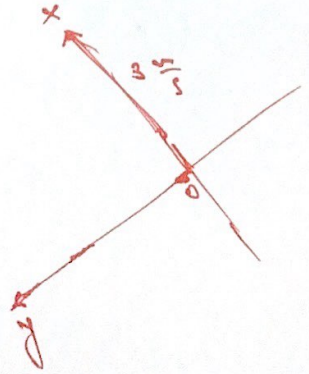
$ma = mg\sin 30° - f$

$f = \mu N = \mu mg\cos 30°$

$a = g\sin 30° - \mu g\cos 30°$

$$a = 2\frac{m}{s^2}$$

ב. נבחר ציר x בכיוון התנועה



$x = x_0 + v_0 t + \frac{1}{2}at^2$

$x = 3t$

ג. נציב את y ונקבל את משוואת הגוף

$y = 2\sqrt{t^2}$

$$y = 2\sqrt{t^2}$$

$t = \frac{x}{3}$

$y = 2\sqrt{\left(\frac{x}{3}\right)^2}$

$y = \frac{2\sqrt{3}}{9}x^2$

$$y = \frac{\sqrt{3}}{9}x^2$$

ד. נציב y ונקבל את משוואת הגוף

$x = x_0 + v_0 t + \frac{1}{2}at^2$

$$\begin{cases} x = 3t + \frac{a}{2}t^2 \\ y = 2\sqrt{t^2} \end{cases}$$

$2\sqrt{} = 2\sqrt{t^2}$

$$t = 1\,s$$

נציב את t ב-x:

$1\sqrt{} = 3\cdot 1 + \frac{a}{2}\cdot 1^2$

$1\sqrt{} = 3 - 0.5a$

$$a = -3\frac{m}{s^2}$$

ה.

$v_x = v_0 + at$

$v_x = 3 - 3\cdot 1$

$$v_x = 0$$

$v_y = v_0 + at$

$v_y = 0 + 2\sqrt{}\cdot 1$

$$v_y = 2\sqrt{}\,\frac{m}{s}$$

1.19

$$\begin{cases} x = x_0 + v_x t \\ y = y_0 + v_{0y} t + \frac{a t^2}{2} \end{cases}$$

$$\begin{cases} 5 = 0 + v_0 \cos 70 \cdot t \\ 12 = 0 + v_0 \sin 70 \cdot t - 5 t^2 \end{cases}$$

$$\begin{cases} 5 = 0.342 v_0 \cdot t \\ 12 = 0.9396 t - 5 t^2 \end{cases}$$

$$v_0 t = 14.619$$

$$12 = 0.9376 \cdot 14.619 - 5 t^2$$

$$12 = 13.733 - 5 t^2$$

$$-1.733 = -5 t^2$$

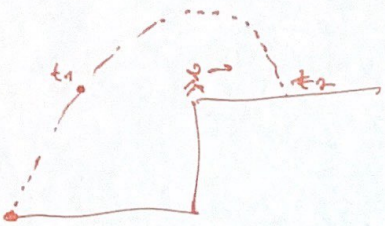
$$0.3476 = t^2$$

$$t = 0.5896 \text{ (s)}$$

$$v_0 t = 14.619$$

$$v_0 = \frac{14.619}{0.5896}$$

$$v_0 = 24.796 \frac{m}{s}$$



$$y = y_0 + v_{0y} t - \frac{1}{2} g t^2$$

$$12 = 0 + 30 \cdot \sin 70 \, t - 5 t^2$$

$$12 = 28.191 t - 5 t^2$$

$$5 t^2 - 28.191 t + 12 = 0$$

$$t_1 = 0.464 \qquad t_2 = 5.174$$

$$x = x_0 + v_0 \cos\theta \cdot t$$

$$x = 0 + 30 \cdot \cos 70 \cdot 0.464$$

$$x = 4.761 \text{ m}$$

$$x = x_0 + v_x t$$

$$x = 4.761 + 30 \cos 70 \, t - 2.5 t^2$$

$$\Delta t = 5.174 - 0.464$$

$$\Delta t = 4.71$$

$x = 4.761 + 30 \cdot \cos 30^\circ \cdot 4.71 + 2.5 (4.71)^2$

$x = 103.543 m$

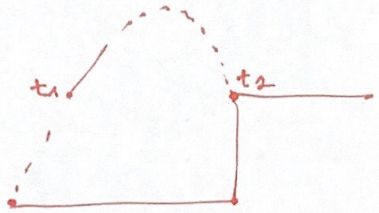
$\Delta x = v \cdot \Delta t$

$103.543 = v \cdot 4.71$

$4.71 v = 103.549$

$v = 21.985 \frac{m}{s}$

(2)



t_2 ... B ... ...

$t_1 = 0.464$ $t_2 = 5.174$

$x_1 = 4.761 m$ $\Delta t = 4.71$

$x = x_0 + v_{0x} t + \frac{a t^2}{2}$

$x = 4.761 + v_0 \cdot \cos 30 \cdot \Delta t + \frac{a \Delta t^2}{2}$

$x = 4.761 + 30 \cdot \cos 30^\circ \cdot 4.71 + \frac{a \cdot 4.71^2}{2}$

$x = 4.761 + 48.327 + 11.092 a$

$a = -4.336 \frac{m}{s^2}$

1.26

א) נבחר ציר x בכיוון התנועה של הכדור ונקודת ראשית בנקודת הזריקה.

- נרשום את משוואת התנועה של הכדור בציר x

$$x = x_0 + v_0 t$$

$$L = 0 + v_0 t_1$$

$$\boxed{L = v_0 t_1}$$

$$\boxed{t_1 = \frac{L}{v_0}}$$

- נרשום את משוואת התנועה של הכדור בציר y

$$y = y_0 + v_{0y} t + \frac{a t^2}{2}$$

$$y = 2.5 + 0 - 5 t_1^2$$

$$y = 2.5 - 5\left(\frac{L}{v_0}\right)^2$$

$$\boxed{y = 2.5 - \frac{5L^2}{v_0^2}}$$

- נרשום את משוואת התנועה של הכדור בציר x בכיוון הקיר

$$x = x_0 + v_0 t$$

$$L = 0 + v_0 t_2$$

$$L = v_0 t_2$$

$$\boxed{t_2 = \frac{L}{v_0}}$$

$$\boxed{t = t_1 + t_2 = \frac{2L}{v_0}}$$

- נרשום את משוואת התנועה של הכדור בציר y בכיוון הקיר

$$y = y_0 + v_{0y} t + \frac{a t^2}{2}$$

$$y = 2.5 + 0 - 5 t_1^2$$

$$y = 2.5 - 5\left(\frac{L}{v_0}\right)^2$$

$$\boxed{y = 2.5 - \frac{5L^2}{v_0^2}}$$

- נרשום את משוואת התנועה של הכדור בציר y בכיוון הקיר

$$y = y_0 + v_{0y} t + \frac{a t^2}{2}$$

$$y = 10 + 0 - 5\left(\frac{L}{v_0}\right)^2$$

$$\boxed{y = 10 - \frac{5L^2}{v_0^2}}$$

- נרשום את משוואת התנועה של הכדור בציר z בכיוון הקיר

$$2.5 - \frac{5L^2}{v_0^2} = 10 - \frac{5L^2}{v_0^2}$$

$$\frac{480}{v_0^2} = 7.5$$

$$v_0^2 = \frac{480}{7.5} = 64$$

$$\boxed{v_0 = 8 \frac{m}{s}}$$

ב)

$$t = \frac{L}{v_0} = \frac{10}{8}$$

$$\boxed{t = 1.25 \, [s]}$$

ג)

$$t_1 = \frac{L}{v_0} = 1.25 \, [s] \qquad t_2 = \frac{L}{v_0} = 0.25 \, [s]$$

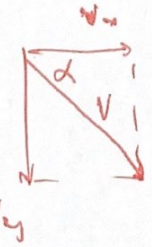
~~$t_2 = \frac{L}{v_0} = \frac{2}{8} = 0.25$~~

$$\boxed{v_x = v_0 = 8 \frac{m}{s}}$$

$$v_y = v_{0y} - 10 t_2$$

$$v_y = 0 - 10 \cdot 0.25$$

$$\boxed{v_y = -2.5 \frac{m}{s}}$$



$$V = \sqrt{V_x^2 + V_y^2}$$

$$V = \sqrt{8^2 + 2.5^2}$$

$$\boxed{V = 8.382 \ \frac{m}{s}}$$

$$\alpha = \tan^{-1}\left(\frac{V_y}{V_x}\right)$$

$$\alpha = \tan^{-1}\left(\frac{2.5}{8}\right)$$

$$\boxed{\alpha = 17.35^0}$$

ב. מהירות הגוף:

$$V_y = V_0 + at$$

$$V_y = 0 - 10 \cdot 1.25$$

$$\boxed{V_y = -12.5 \ \frac{m}{s}}$$

$V_x = 8$

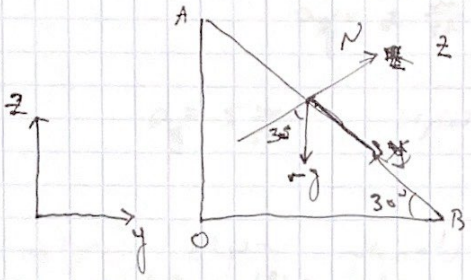
ג. נמצא את גובה הנקודה הגבוהה

$$y = y_0 + V_0 t + \frac{at^2}{2}$$

$$y = 10 + 0 - 5 \cdot (1.25)^2$$

$$\boxed{y = 2.1875 \ m}$$

1.29





$$\Sigma F_z = 0$$

$$N - mg\cos 30^\circ = 0$$

$$\Sigma F_y = ma$$

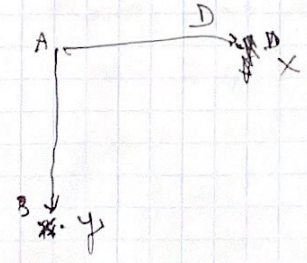
$$mg\sin\theta = ma$$

$$\boxed{a = g\sin\theta}$$

$$a = g\sin 30^\circ$$

$$a = 0.5g$$

$$\boxed{a = 5\ \frac{m}{s^2}}$$



ציר X

$a_x = 2 \frac{m}{s^2}$

ציר Y

$a_y = 5 \frac{m}{s^2}$

(שים לב: בכיוון ציר Y יש תאוצה קבועה שאינה אפס, ולכן הגוף לא יישאר במנוחה לאורך ציר זה.)

ציר Y

$x_0 = 0,\ v_{0x} = 0,\ a_x = 5 \frac{m}{s^2}$ $\quad y = y_0 + v_{0x}t + \frac{1}{2}a_y \cdot t^2$

$\boxed{y = 2.5t^2}$

ציר X

$y_0 = 0,\ v_{0y} = 0\ \ a_y = 2 \frac{m}{s^2}$ $\quad x = x_0 + v_{0y} \cdot t + \frac{1}{2}a_y t^2$

$\boxed{x = t^2}$

כדי למצוא את משוואת המסלול, נציב את הביטוי עבור t במשוואת התנועה בציר Y:

$\frac{y}{x} = 2.5$

$\boxed{y = 2.5x}$

משוואת המסלול היא משוואת ישר העובר בראשית הצירים.

(ב) כאשר $x = 10m$ הגוף נמצא במרחק

$y = 2.5t^2$

$10 = 2.5t^2$

$t^2 = 4$

$\boxed{t = 2s}$

(2)

$$\begin{cases} v_x = v_{0x} + a_x t \\ v_y = v_{0y} + a_y t \end{cases}$$

$$\begin{cases} v_x = 5t \\ v_y = 2t \end{cases}$$

$t = 2(s)$

$v_x = 5 \cdot 2 = 10$

$v_y = 2 \cdot 2 = 4$

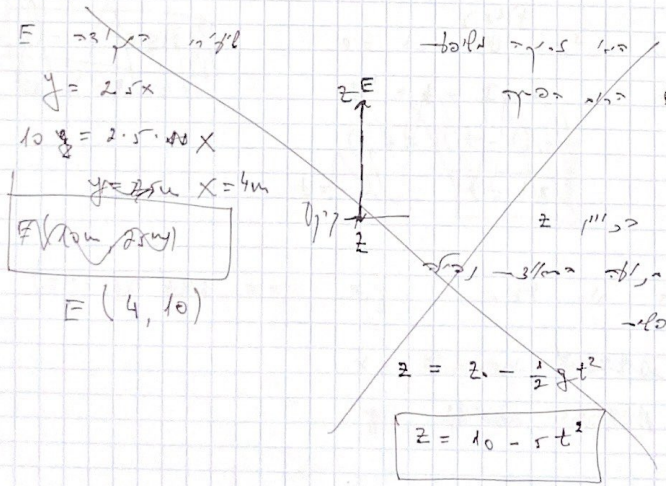
$$v_0 = 10\ \tfrac{m}{s}$$
$$v_y = 4\ \tfrac{m}{s}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{4^2 + 10^2}$$

$$v = \sqrt{116}$$

$$v = 10.77\ \tfrac{m}{s}$$

(3)



$y = 2.5x$

$10 = 2.5 \cdot x$

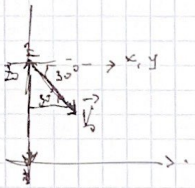
$x = 4m$

$E(4, 10)$

$$z = z_0 - \frac{1}{2} g t^2$$

$$z = 10 - 5t^2$$

30° -ის ქვეშ ტყორცნეს ქვა სიჩქარით E ველში



$$\begin{cases} x = x_0 + v_{0x} \cdot t \\ y = y_0 + v_{0y} \cdot t \\ z = z_0 + v_{0z} \cdot t + \frac{1}{2} a_z t^2 \end{cases}$$

$x = 4 + 4t$

$y = 10 + 10t$

$z =$

$x_0 = 4$ მ

$v_{0x} = 4 \cdot \cos 30°$

$y_0 = 10$

$v_{0y} = 10 \cdot \cos 30°$

$z_0 = 10$ მ

$v_{0z} = 10 \cdot \sin 30° =$

$10 \cdot 77 \cdot 0.5 = 5.385$ მ/წმ

$$\begin{cases} x = 4 + 3.464t \\ y = 10 + 8.66t \\ z = 10 - 5.385t - 5t^2 \end{cases}$$

$z = 0$ მიწაზე დაცემის დრო

$t_1 = 0.925$ წმ

$t_2 = -2.16$

$0 = 10 - 5.385t - 5t^2$ $\quad t_1 = 0.925$ წმ

$t^2 + t - 2 = 0$ $\quad t_2 = -2$

$(t+2)(t-1) = 0$

$t = -2$ $\quad t = 1$ წმ

$t = 0.925$ წმ-ში x, y კოორდინატები

$x = 4 + 3.464 \cdot 0.925 = 7.142$ მ

$y = 10 + 8.66 \cdot 0.925 = 17.851$ მ

$$d = \sqrt{(x-5)^2 + (y-5)^2}$$

$$d = \sqrt{(7.464-5)^2 + (13.66-5)^2}$$

$$d = \sqrt{(7.142-5)^2 + (17.355-5)^2}$$

$$\boxed{d = 13.03\ m}$$